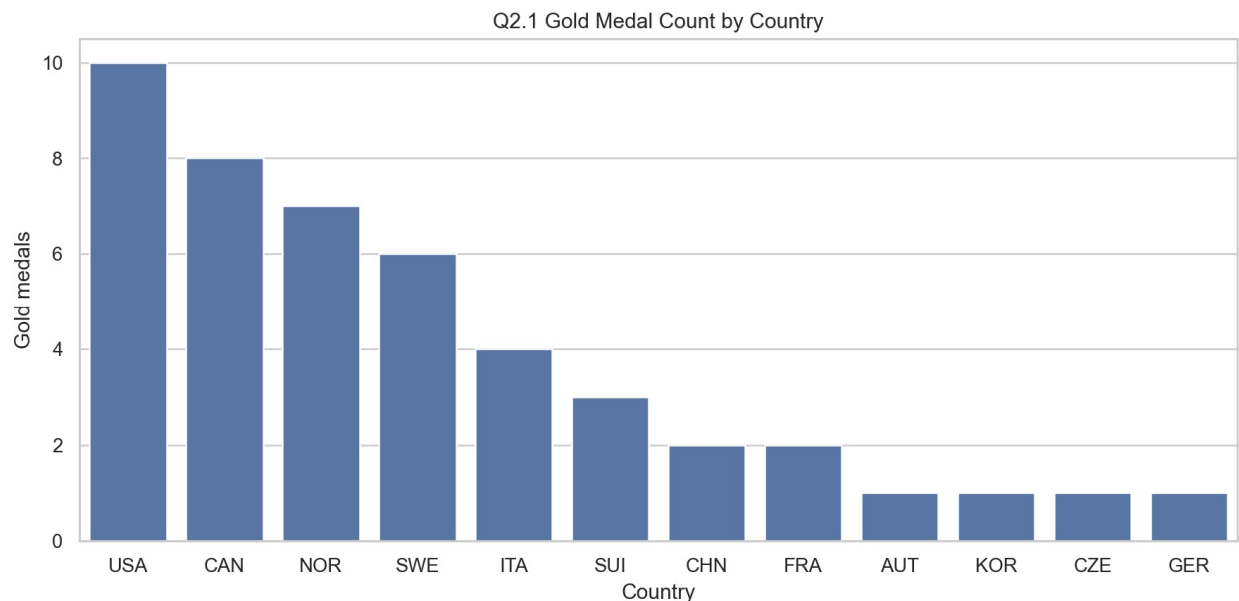


Task 2: Exploratory Data Analysis

Introduction

In this section, we analyzed the dataset to understand what drives medal performance in the Winter Olympics. We used visualizations to examine athlete metrics, training load, and country factors with respect to success. Key questions are as follows: the differences between the United States and Canada, the profile of gold medallists, and how sport, age, experience, and physiology relate to success. We used the data to identify clear patterns and support each finding with evidence. Each part of the analysis answers a specific question and shows how different factors relate to medal outcomes. Together, these results explain what separates top athletes and how performance varies across sports and countries.

Q2.1: USA vs Canada

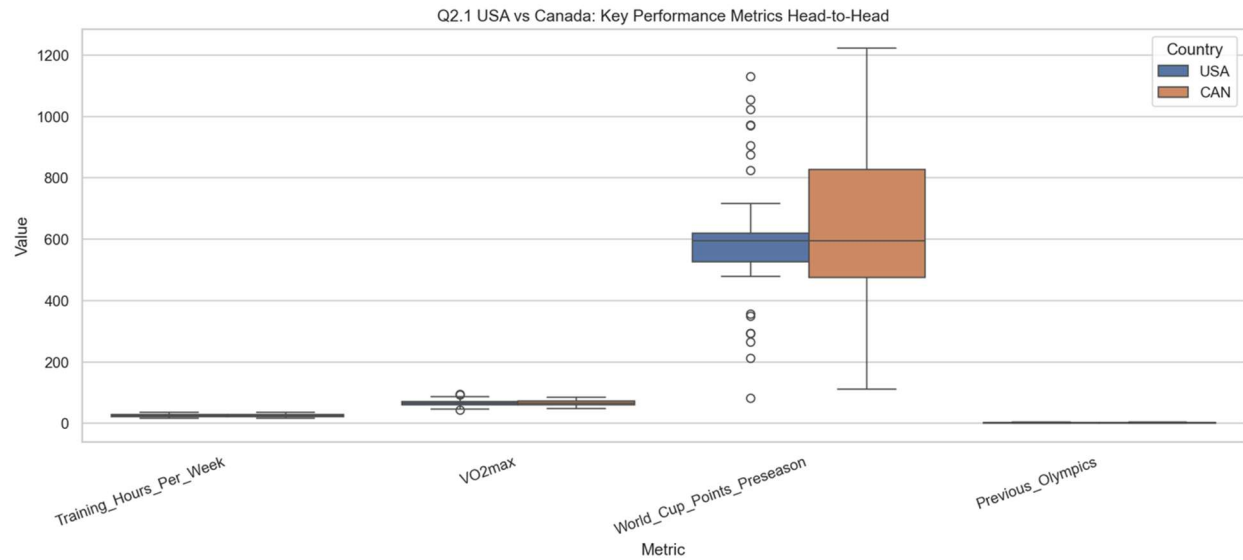


Title: Gold Medal Count by Country (USA vs Canada)

X-axis: Country

Y-axis: Number of Gold Medals

Interpretation: The USA won 10 gold medals, while Canada won 8. This shows a clear advantage for the USA at the top of the podium. The gap suggests that the USA converted performance into gold medals more effectively than Canada.



Title: Comparison of Key Performance Metrics (USA vs Canada)

X-axis: Performance Metric

Y-axis: Value

Interpretation: USA athletes showed slightly higher training hours and VO2max values than Canadian athletes. Canada showed a wider spread in World Cup points, with some high-performing outliers. This suggests that the USA had more consistent performance, while Canada relied on a few standout athletes.



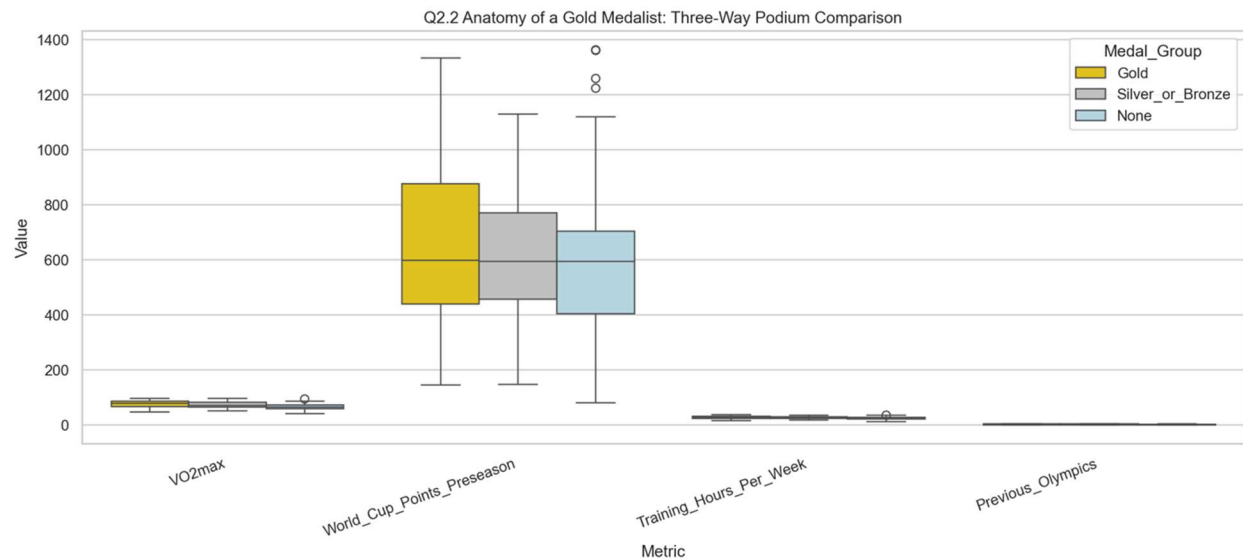
Title: Athlete Distribution by Sport (USA vs Canada)

X-axis: Country

Y-axis: Sport

Interpretation: The USA had more athletes in alpine skiing, cross-country, and figure skating, while Canada had higher representation in freestyle skiing, snowboard, and speed skating. This shows that each country focused on different sport portfolios. The USA's broader spread across sports may have increased its chances of winning more gold medals.

Q2.2: Gold vs Others

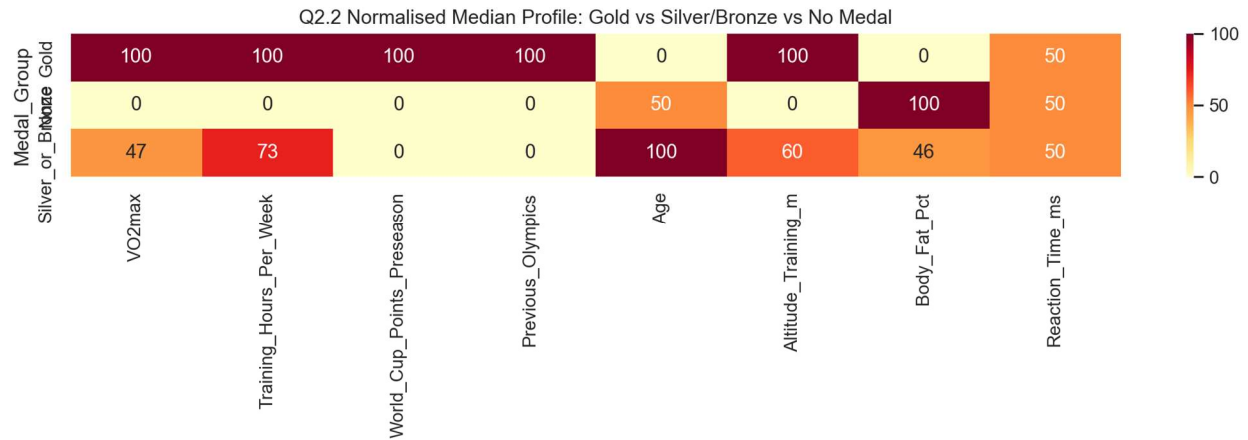


Title: Comparison of Key Performance Metrics by Medal Group (Gold vs Silver/Bronze vs No Medal)

X-axis: Performance Metric

Y-axis: Value

Interpretation: Gold medallists showed higher VO2max, higher World Cup points, and slightly higher training hours compared to other groups. The gap between gold and silver/bronze was smaller than the gap between medallists and non-medallists. This suggests that reaching the podium depends on strong performance, while winning gold requires a smaller additional edge.



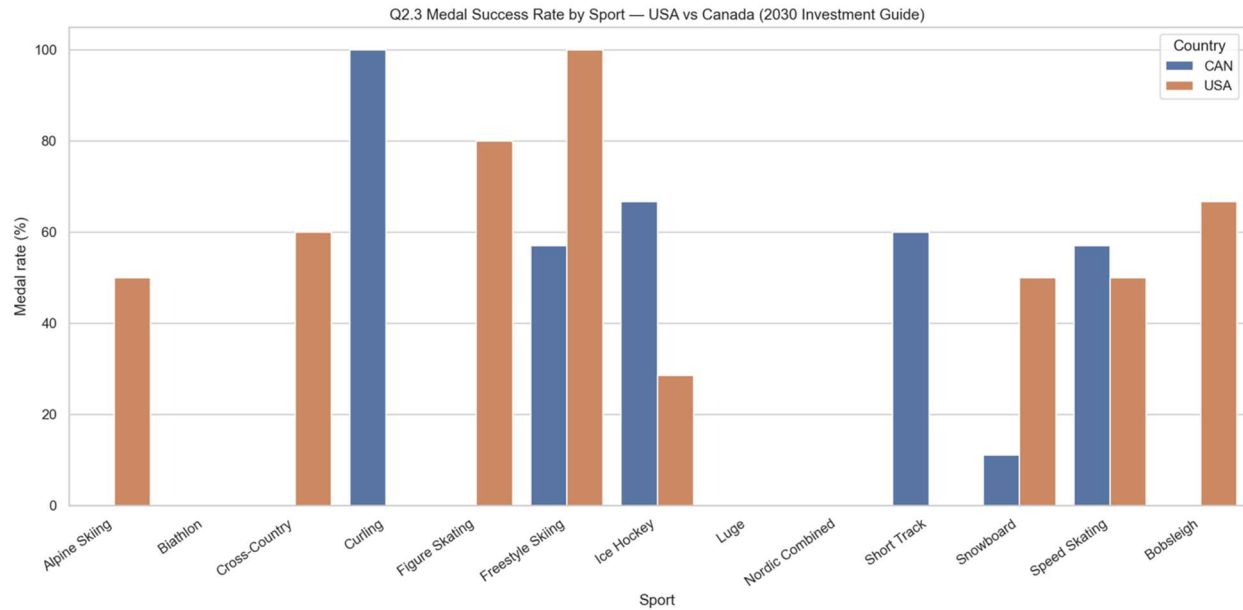
Title: Normalized Median Performance Profile (Gold vs Silver/Bronze vs No Medal)

X-axis: Performance Variables

Y-axis: Medal Group

Interpretation: Gold medallists ranked highest across most key variables, including VO2max, training hours, and World Cup points. Silver and bronze medallists showed moderate values, while non-medallists scored lower in most categories. Age and reaction time showed less clear separation, which suggests they are weaker predictors of gold-level performance.

Q2.3: Sport Strategy



Title: Medal Success Rate by Sport (USA vs Canada)

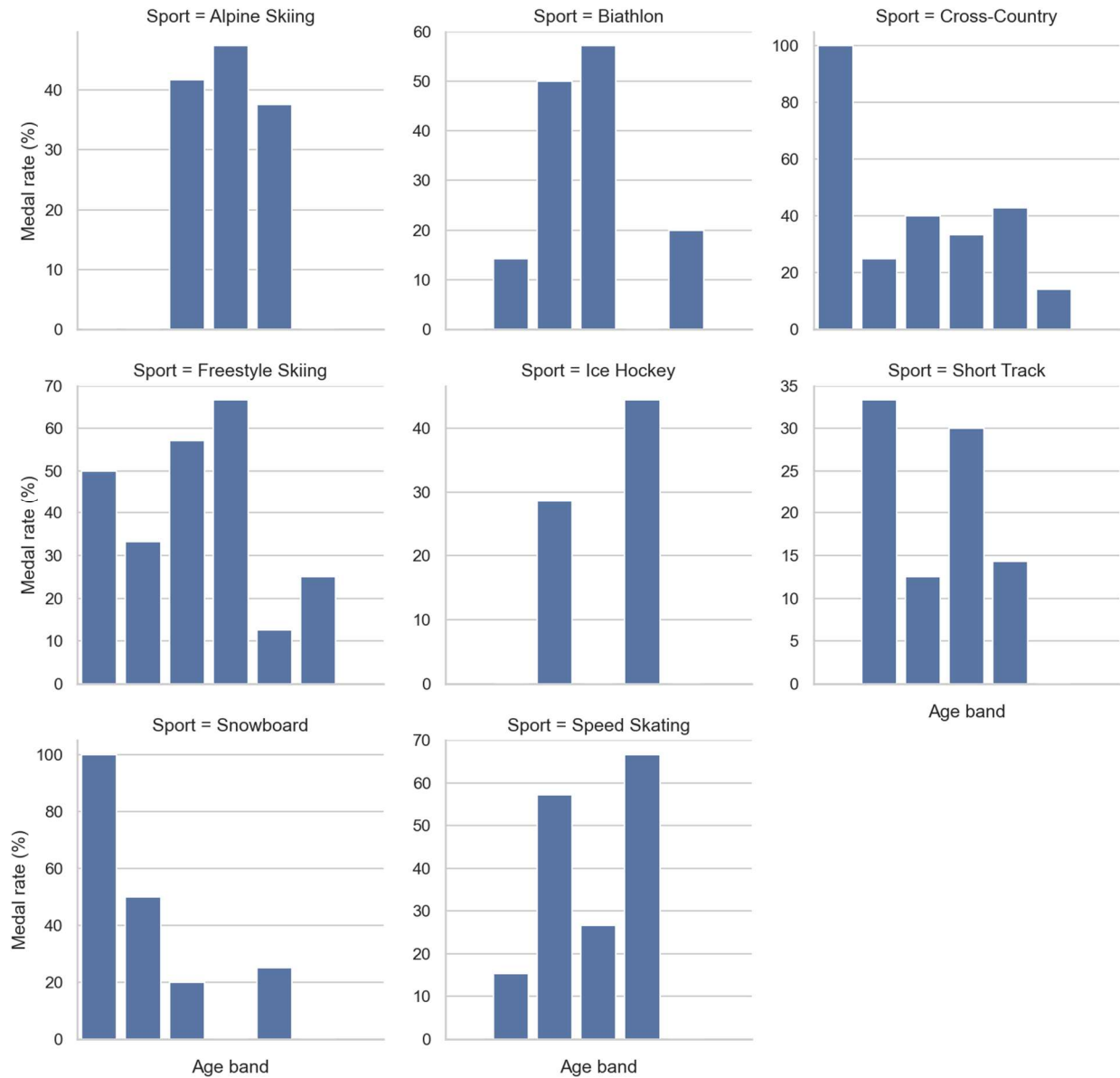
X-axis: Sport

Y-axis: Medal Success Rate (%)

Interpretation: The USA showed high medal success rates in freestyle skiing, figure skating, and bobsleigh, with several sports reaching close to or at 100%. Canada showed strong performance in curling, ice hockey, and short track, where medal rates were consistently higher than those of the USA. In contrast, both countries showed weaker performance in sports such as luge and Nordic combined, where medal rates were low or absent.

Q2.4: Age & Experience

Q2.4 Age Peak Window by Sport



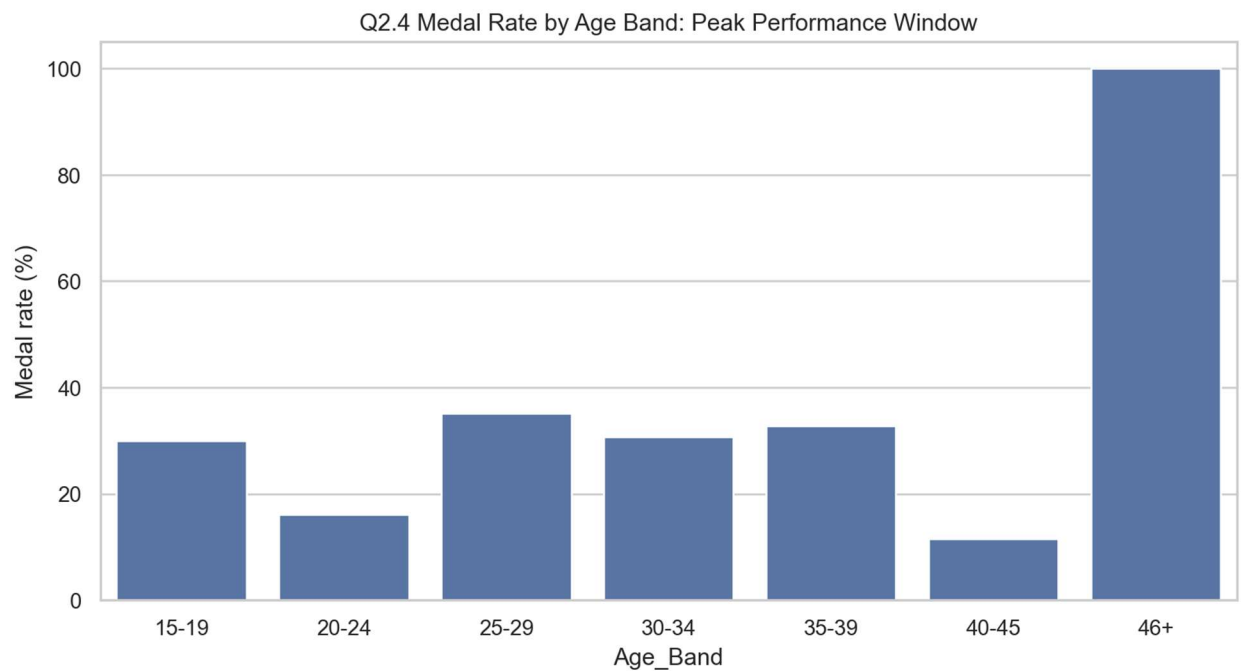
Title: Medal Rate by Age Band Across Sports

X-axis: Age Band

Y-axis: Medal Rate (%)

Interpretation: Peak performance varied across sports, but most disciplines showed higher medal rates in the 25-34 age range. Some sports, such as freestyle skiing and speed

skating, showed strong performance in the late 20s and early 30s. This suggests that athletes often reach their peak after several years of development and competition experience.

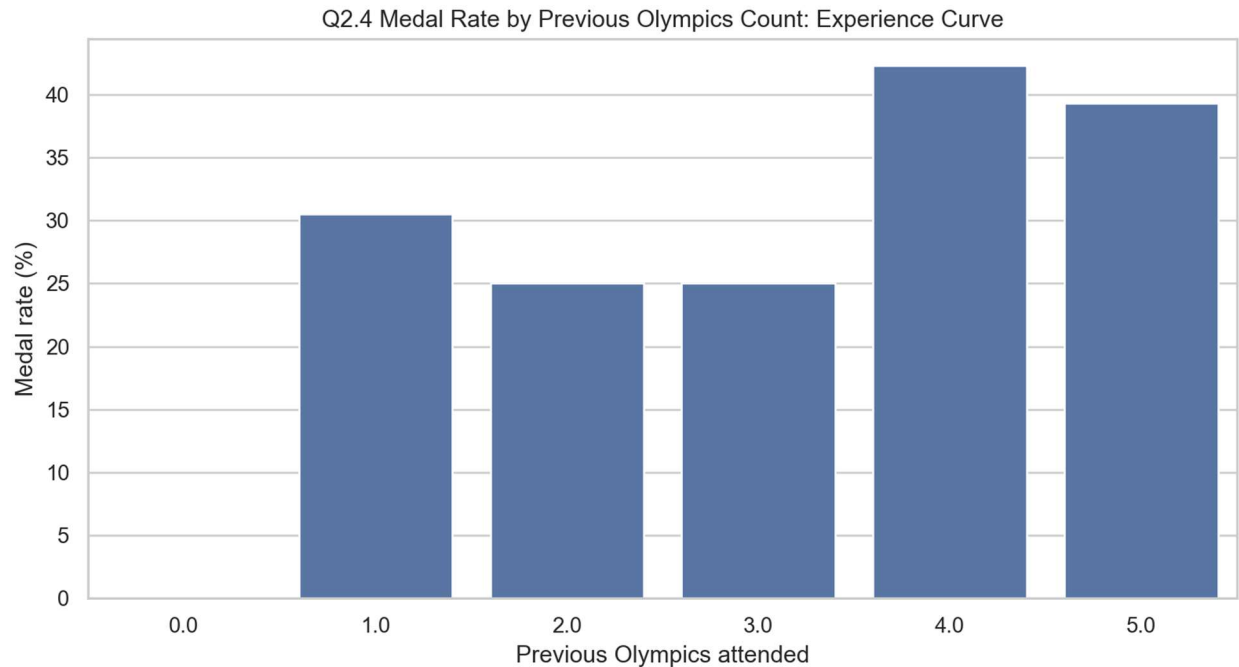


Title: Medal Rate by Age Band (All Athletes)

X-axis: Age Band

Y-axis: Medal Rate (%)

Interpretation: The highest medal rates appeared in the 25-29 and 30-34 age groups. Younger athletes (20-24) showed lower medal rates, while older groups (35-39) remained competitive but with slightly lower performance. The 46+ group showed a high rate, but this likely reflects a small number of exceptional cases rather than a general trend.



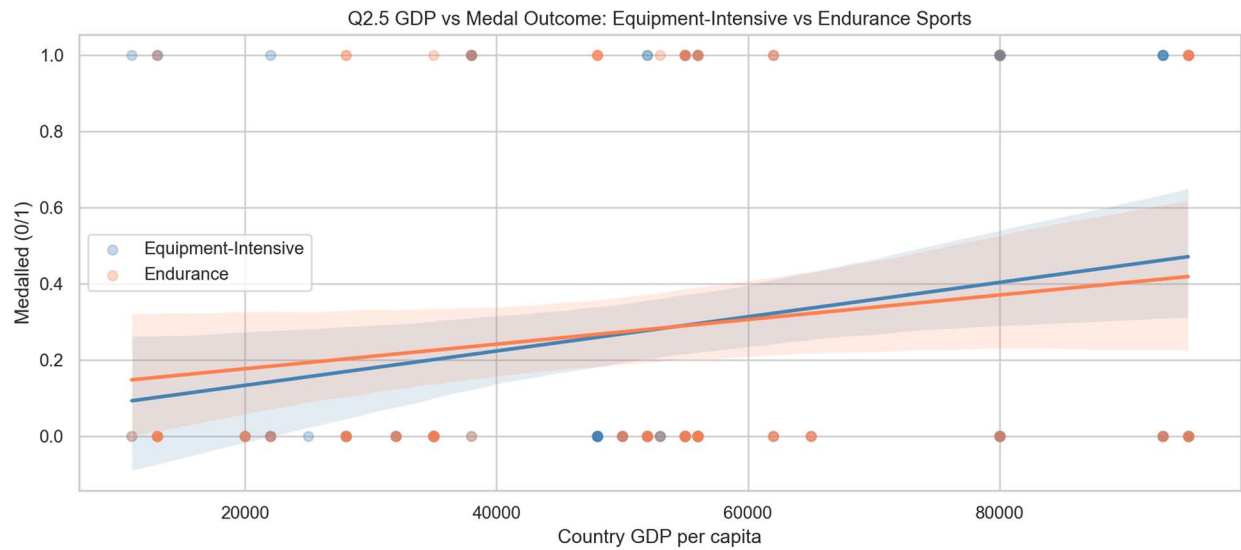
Title: Medal Rate by Number of Previous Olympics

X-axis: Number of Previous Olympics

Y-axis: Medal Rate (%)

Interpretation: Athletes with more Olympic experience showed higher medal rates. Medal success increased from first-time participants to athletes with multiple appearances, with a peak around four Olympic participations. This suggests that experience plays a strong role in performance at the Olympic level.

Q2.5: GDP & Tradition

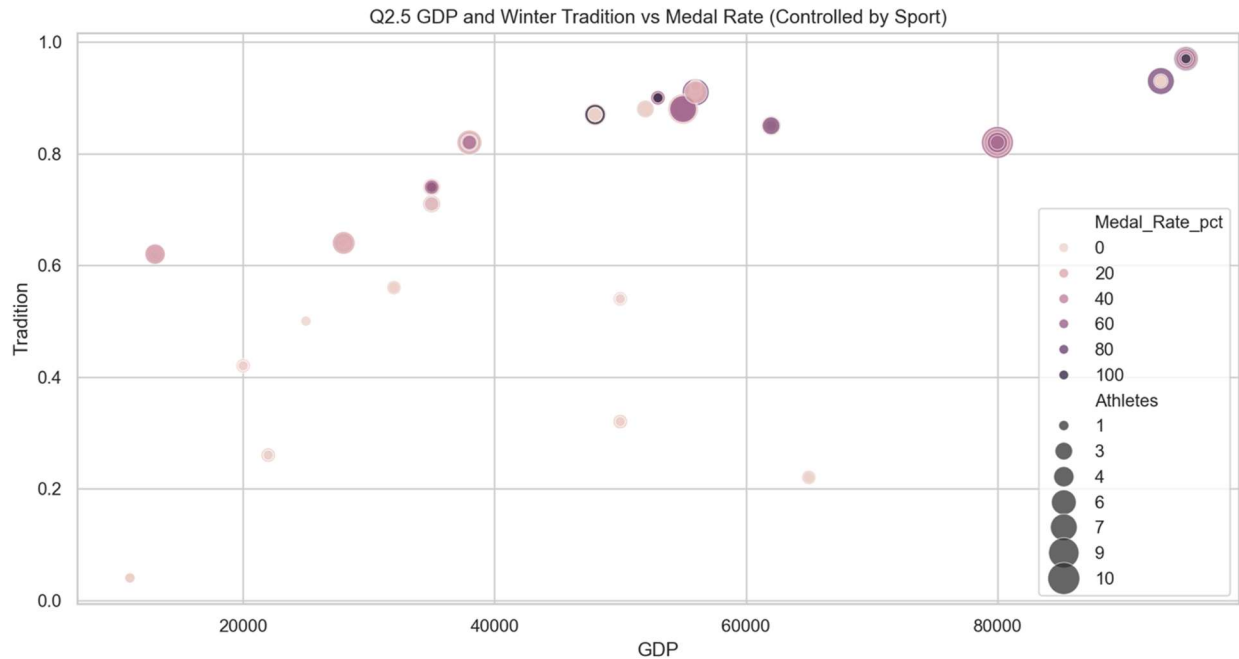


Title: GDP per Capita vs Medal Outcome (Equipment-Intensive vs Endurance Sports)

X-axis: Country GDP per Capita

Y-axis: Medal Outcome (0 = No Medal, 1 = Medal)

Interpretation: Medal probability increased as GDP per capita increased for both sport types. The relationship appeared slightly stronger in equipment-intensive sports than in endurance sports. This suggests that financial resources may provide a greater advantage in sports that require advanced equipment and infrastructure.

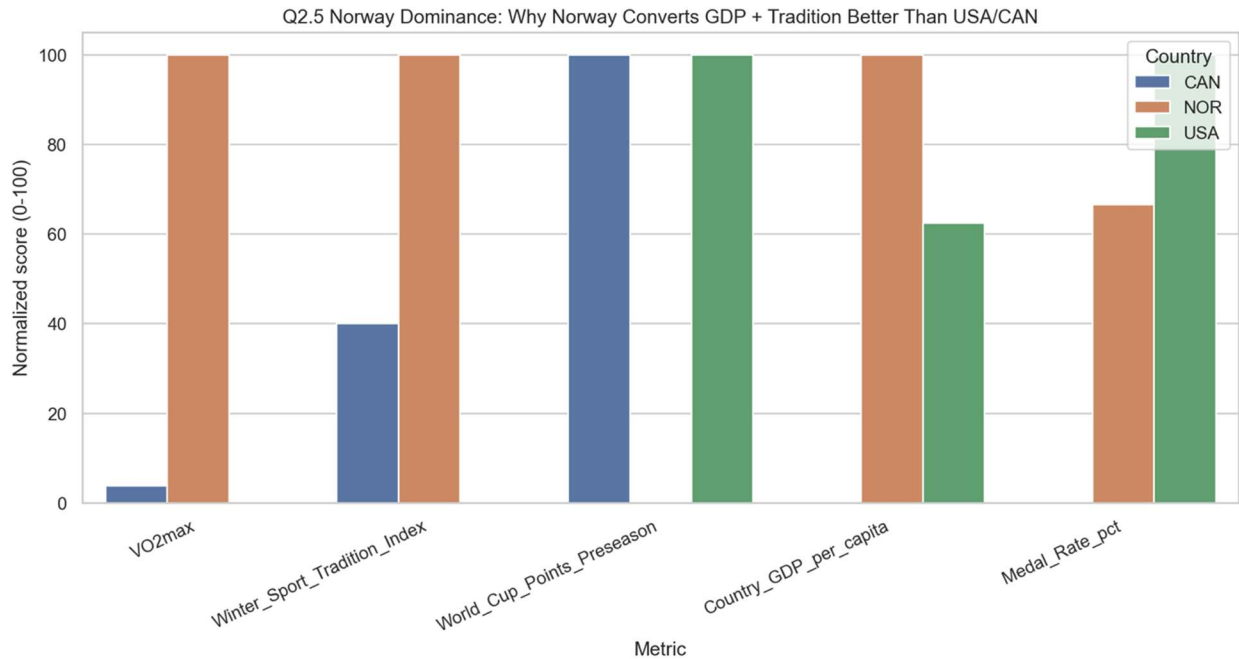


Title: GDP and Winter Sport Tradition vs Medal Rate

X-axis: GDP per Capita

Y-axis: Winter Sport Tradition Index

Interpretation: Countries with both high GDP and strong winter sport tradition achieved higher medal rates. High-performing countries clustered in the upper-right region of the plot. This indicates that success depends on both economic resources and cultural investment, rather than GDP alone.



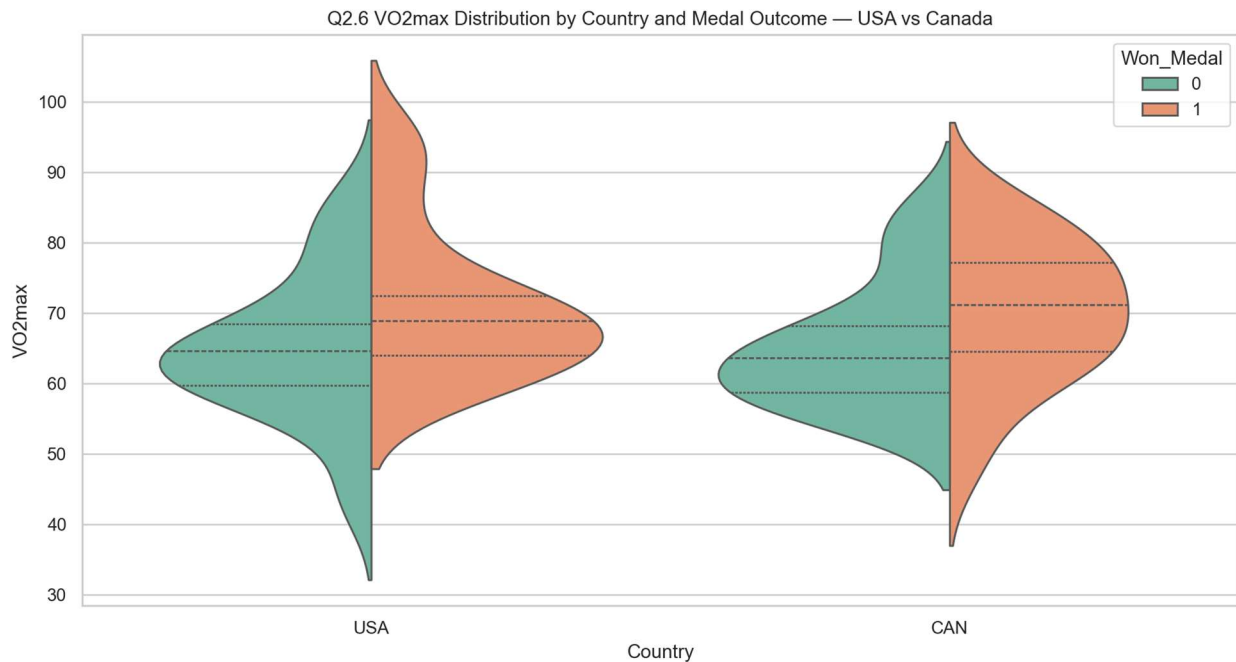
Title: Comparison of Key Performance Factors (Norway vs USA vs Canada)

X-axis: Performance Metric

Y-axis: Normalized Score (0–100)

Interpretation: Norway showed consistently high scores across all variables, including VO2max, tradition index, and GDP. The USA and Canada showed strong GDP but lower scores in tradition and efficiency metrics. This suggests that Norway converts its resources and tradition into performance more effectively than the USA and Canada.

Q2.6: Physiological Profile

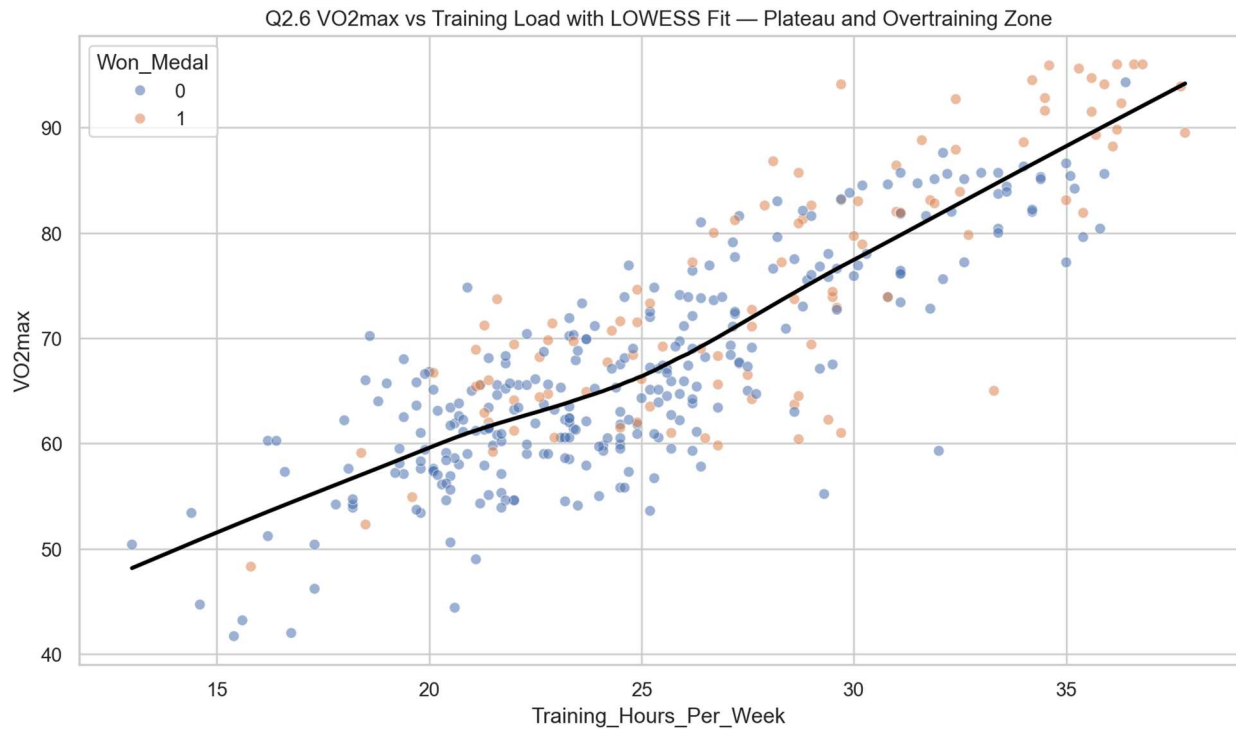


Title: VO2max Distribution by Country and Medal Outcome (USA vs Canada)

X-axis: Country

Y-axis: VO2max (ml/kg/min)

Interpretation: Athletes who won medals showed higher VO2max values than non-medallists in both countries. Canadian medallists showed slightly higher VO2max levels than USA medallists, while non-medallists from both countries clustered at lower values. This indicates that strong aerobic capacity is a key factor for reaching the podium.



Title: VO2max vs Training Hours per Week with Trend Line

X-axis: Training Hours per Week

Y-axis: VO2max (ml/kg/min)

Interpretation: VO2max increased as training hours increased, showing a strong positive relationship. However, the curve showed signs of flattening at higher training loads, which suggests diminishing returns beyond a certain point. This indicates that very high training volumes may not lead to proportional performance gains.